

SOFTWARE DEVELOPMENT PLAN (SDP)

for the

Task Force Management System (TFMS)

Contract No: MSE-FT31-2025

Prepared For:

Universiti Teknologi Malaysia (UTM)

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1 - SCOPE

This section shall be divided into the following paragraphs.

1.1 Identification

This software development plan establishes the plan for the development of Task Force Management System for Universiti Teknologi Malaysia (UTM). Below are the details of the system to be developed:

System name: Task Force Management System
System abbreviation:
System identification number: TFMS-CODEForce

CSCI name:
CSCI abbreviation:
CSCI identification number:

1.2 System overview

Purpose of the system: The Task Force Monitoring System (1.0) provides a secure, centralised digital platform to manage lecturer task force assignments. It replaces the current manual, spreadsheet-based approach by supporting transparent workload balancing, streamlined membership updates, automated weightage computation, approval workflows, and reliable reporting. The system ensures accuracy, fairness, and traceability in task force management across the Faculty of Artificial Intelligence (FAI). Looking ahead, the TFMS is designed with future expansion in mind, allowing scalability to additional faculties, campuses, or institutes without major redesign. It also anticipates integration with other university systems, such as single sign-on (SSO), human resource systems, and academic records, ensuring interoperability and alignment with institutional governance policies.

1.3 Document overview

The Software Development Plan (SDP) describes the contractor's estimates for achieving the development of the Task Force Coordination and Monitoring. This document is used to fully inform UTM of the organization set up for software development and of the methods applied. This document allows the Government to control the procedures, management and means implemented for software development. The following table describes the contents of the SDP:

Table 1: SDP Contents

CHAPTER	CONTENTS	DESCRIPTION
1	Scope	Describe the identification of the CSCI, system overview and this document overview.
2	Referenced Documents	Describes the listing of Government documents and Non-Government documents that are used for reference in this project.
3	Software Development Management	Describes the planning associated with software development management activities.

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4	Software Engineering	Describes the planning associated with software engineering activities.
5	Formal Qualification Testing	Describes the planning associated with formal qualification activities.
6	Software Product Evaluation	Describes the planning associated with software product evaluation activities.
7	Software Configuration Management	Describes configuration management consists in applying technical and administrative rules and supervision
8	Other Software Development Functions	Not Applicable
9	Notes	List of abbreviations in this document.

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2 - REFERENCED DOCUMENTS

2.1. Government documents

- | | | |
|-----|---------------|---|
| [1] | DoD-Std-2167A | Defense System Software Development |
| [2] | DoD-Std-2168 | Defense System Software Quality Program |
| [3] | MIL-Std-1521B | Technical reviews and audits |

2.2. Non-Government documents

- | | | |
|-----|-------------------------------|--|
| [1] | DI ² (Man_Doc_001) | General organization of the project |
| [2] | DI ² (Man_Doc_002) | Presentation of the project |
| [3] | DI ² (Man_Doc_003) | Software Specification Review |
| [4] | DI ² (Man_Doc_004) | Preliminary Design Review |
| [5] | DI ² (Man_Doc_005) | Test Readiness Review |
| | DI ² (Man_Doc_006) | User Requirements Specification for Taskforce Monitoring System
() |
| [6] | RDL-105 | Conduct of reviews and audits guide |
| [7] | Guideline UML-1 | The Unified Modeling Language User Guide (Addison Wesley) |
| [8] | Guideline UML-2 | The Unified Software Development Process (Addison Wesley) |
| [9] | PM Guideline | Guide to the Software Engineering Body of Knowledge |

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3 - SOFTWARE DEVELOPMENT MANAGEMENT

This section shall be divided into the following subparagraphs to describe the planning associated with software development management activities.

3.1 Project organization and resources

This section shall be divided into the following subparagraphs to describe the project organization and the project resources of the contractor.

3.1.1 Contractor facilities

Table 2 below is the description of the contractor's facilities to be used for the contracted effort.

Table 2: Contractors' Equipment

No.	Product	Version
1.	Windows Sandbox	Latest version or compatible with Window OS version
2.	Testing Frameworks/Tools (Selenium, Apache JMeter, Postman, Chrome Developer Tools, Wireshark)	Latest version or compatible with Window OS version
3.	File Sharing (OneDrive, GoogleDrive)	Latest version or compatible with Window OS version
4.	Project Management Suite (Microsoft Project)	Latest version or compatible with Window OS version
5.	Operating System (Windows 11 or Linux)	Latest version or compatible with Window OS version
6.	Design Tools (Draw.io, Lucid.app)	Latest version or compatible with Window OS version
7.	Git	Latest version or compatible with Window OS version
8.	Visual Studio Code	Latest version or compatible with Window OS version
9.	Microsoft Team	Latest version or compatible with Window OS version

3.1.2 - Client furnished equipment, software and services

Table 3 below is the description of the client's facilities to be used for the contracted effort.

Table 3: Client's Furnished Equipment (UTM)

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No.	Product	Version
1.	Workstation for Project Team with office facilities (printers, shredder, etc.)	Not applicable
2.	Development, Staging and Production Server, included with Windows and Linux OS	Latest version or compatible with Window OS version

3.1.3 - Organizational structure

Below is an overview of the contractor's software project Organizational Breakdown Structure.

Figure 1 shows the overview of the Organizational Breakdown Structure (OBS).

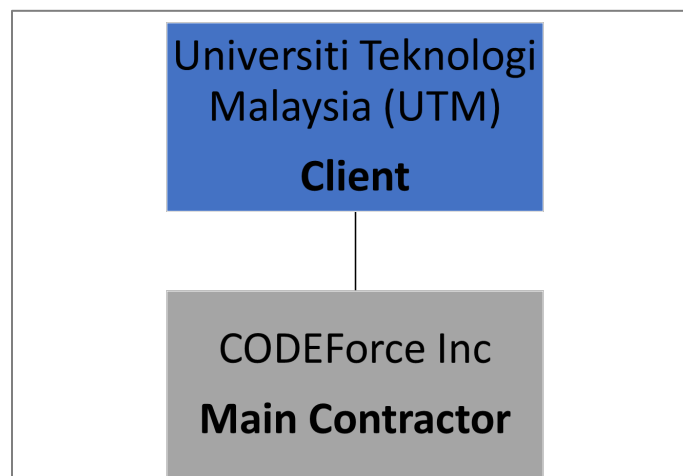


Figure 1: Organizational Breakdown Structure (OBS)

The roles and responsibility for OBS shows as table 4 below:

Table 4: Roles and Responsibility

No.	Organization	Roles and Responsibility
1.	UTM	The client
2.	CODEForce Inc	The prime contractor and responsible for the system integration

3.1.4 - Personnel

Table 5 below are the total numbers of personnel necessary to complete the software development project.

Table 5: Personnel

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No	Roles	Number
1	Software Work Package Manager	1
2	Quality Manager	1
3	Configuration Manager	1
4	Developer	2
5.	Tester	2

3.2 Schedule and milestones

This paragraph shall be divided into the following subparagraphs.

3.2.1 - Activities

The figure 2 shows the Work Breakdown Structure (WBS) for this project:

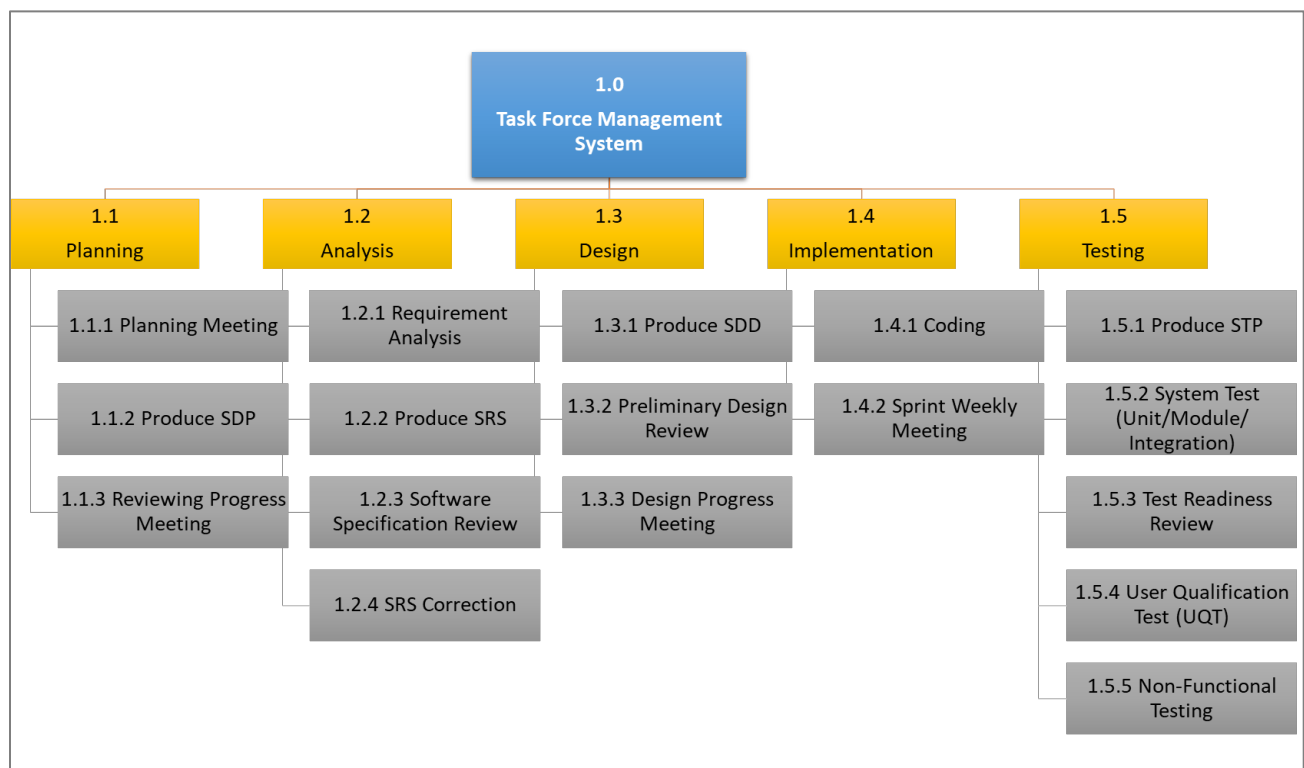


Figure 2: Work Breakdown Structure (WBS)

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The figure 3 depicts the Gantt chart for this project listing all the activities mentioned in the WBS.

Phase	Task ID	Task Name	Duration	Dependencies	December '25					January '26			
1.0 Project	1	Task Force Management System	—	—	W1	W2	W3	W4	W5	W6	W7	W8	W9
Planning	1.1	Planning	—	—									
	1.1.1	Planning Meeting	2 weeks	—									
	1.1.2	Produce SDP	2 weeks	1.1.1									
	1.1.3	Reviewing Progress Meeting	9 weeks	1.1.2									
Analysis	1.2	Analysis	—	—									
	1.2.1	Requirement Analysis	2 weeks	1.1.2									
	1.2.2	Produce SRS	2 weeks	1.2.1									
	1.2.3	Software Specification Review (SSR)	1 day	1.2.2									
	1.2.4	SSR Correction	2 weeks	1.2.3									
Design	1.3	Design	—	—									
	1.3.1	Produce SDD	2 weeks	1.2.3									
	1.3.2	Preliminary Design Review (PDR)	1 day	1.3.1									
	1.3.3	Design Progress Meeting	2 weeks	1.3.2									
Implementation	1.4	Implementation	—	—									
	1.4.1	Coding	5 weeks	1.3.2									
	1.4.2	Sprint Weekly Meeting	5 weeks	1.4.1									
Testing	1.5	Testing	—	—									
	1.5.1	Produce STP	2 weeks	1.2.1									
	1.5.2	Unit/Module Testing	2 weeks	1.4.1									
	1.5.3	Component Integration & System Testing	2 weeks	1.5.2									
	1.5.4	Test Readiness Review (TRR)	1 day	1.5.3									
	1.5.5	User Qualification Test (UAT)	3 days	1.5.4									
	1.5.6	Non-Functional Testing	2 days	1.5.5									

Figure 3: Gantt Chart

3.2.2 - Activity network

The figure 4 depicts the PERT (Program Evaluation and Review Technique) chart for this project, listing all the activities mentioned in the Work Breakdown Structure (WBS). The chart visually represents the project's timeline, emphasizing task sequencing, duration, and dependencies.

The arrows between nodes indicate the dependencies between activities, ensuring a logical progression of tasks throughout the project lifecycle. Each node represents a specific activity, labeled with its name and the estimated duration in days or weeks, allowing project stakeholders to understand the required timeline for each task. The flowchart depicted in a figure 4 below:

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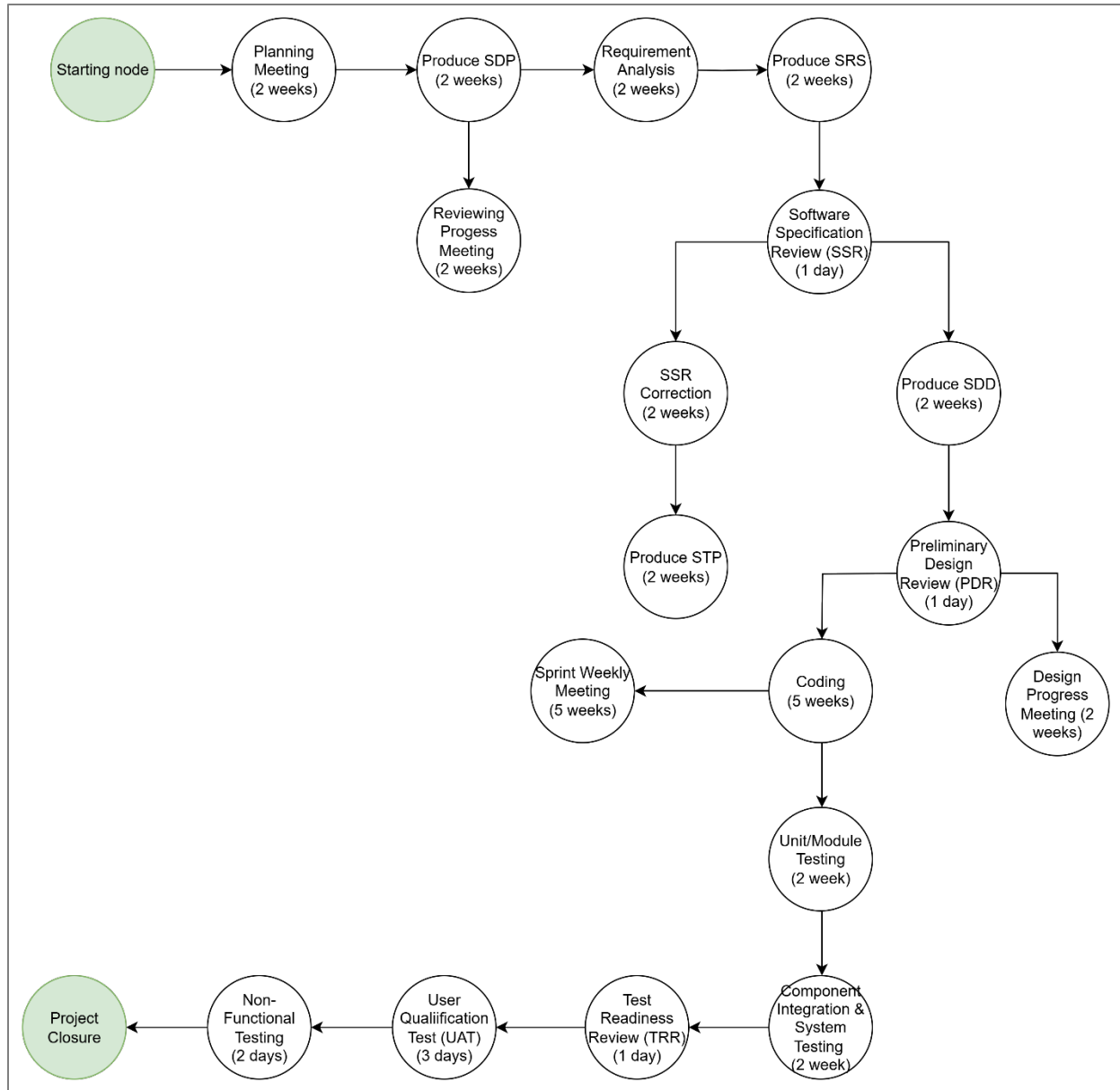


Figure 4: PERT Chart

3.2.3 - Source identification

Table 6 below are the required resources, the need date and availability of each resource item

Table 6: Source Identification

No.	Resource	Date Needed	Availability
1.	Development, Staging and Production Server,	Before project kick off	Available

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	come with Windows OS		
2.	Windows Sandbox	A week before security testing start	Available
3.	Testing Frameworks/Tools (Selenium, Apache JMeter, Postman, Chrome Developer Tools, Wireshark)	A week before coding start	Available
4.	File Sharing (OneDrive, GoogleDrive)	Before project kick off	Available
5.	Project Management Suite (Microsoft Project)	Before project kick off	Available
6.	Operating System (Windows 11 or Linux)	Before project kick off	Available
7.	Design Tools (Draw.io, Lucid.app)	A week before requirement start	Available
8.	Git	A week before coding start	Available
9.	Visual Studio Code	A week before coding start	Available
10.	Microsoft Team	A week before coding start	Available

3.3 Risk management

Risk management in this project includes the risk assessment (risk identifying, risk analysis, risk evaluation) and risk treatment process and compliant with ISO31000:2018. This process will be supported by the other procedures such as monitoring and review, recording and reporting, communication and consultation in order to reduce the possibility and limit the negative impact of a risk factor becoming a problem.

Likert Scale, 1-Low, 3-Medium and 5-High has been using to indicate the level of the Possibility and Loss

The Impact Classification Matrix used to identify the impact of risk the project as table 7 below:

Table 7: Impact Classification Matrix

Possibility/Loss	1	3	5
1	Low (1)	Low (3)	Low (5)
3	Low (3)	Medium (9)	High (15)
5	Low (5)	High (15)	Very High (25)

Below are the identified risks categorized into project risks and technical risks

Project Risk:

Project risk refers to the potential issues or uncertainties that can impact the overall success of the project, including scope, schedule, budget, and resources. Table 8 below show the project risk identified for this project

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Table 8: Project Risk

Risk ID	Risk	Impact = Possibility x Loss	Key event	Mitigation	Person In Charge
RP01	Team member's competencies/skills	15	Progress doesn't meet KPI, Task Delayed	Regular Training, Cross Training, Mentoring and Coaching, Continuous Earning Culture	Software Work Package Manager
RP02	Activities /documents submission delay	15	Submission overdue, progress delay	Review Schedule, Document Templates, Regular Check-Ins, Set Milestones	Software Work Package Manager
RP03	Human resource limitation (shortage team members)	15	Burn out, Progress doesn't meet KPI	Workload Management, Remote Work Options, Collaborative Tools, Cross-Training, Clear Communication	Software Work Package Manager
RP04	Miscommunication/ Communication Channel Breakdown	9	Misunderstanding Task Requirement, Multiple Queries	Multiple Channels, Clear Communication Protocols, Regular Updates, Designated Point of Contact, Documentation	Software Work Package Manager
RP05	Team member's attendance	9	Attendance does not meet percentage required	Clear Attendance Policies, Flexible Work Arrangements, Attendance Tracking, Backup Plans	Software Work Package Manager
RP06	Electricity and Network Connection Issues	9	Report, Slow respond or no respond during session	Uninterruptible Power Supply (UPS), Backup Generators, Data Backup, Remote Access	Software Work Package Manager
RP07	Data Privacy Concerns	9	Regulatory Changes, Data Breach	Compliance with Security Regulations, Data Retention Policies, Data Minimization, Data Encryption, Access Control	Quality Manager

Technical Risks:

Technical risk refers to the potential issues or uncertainties related to the technical aspects of a project, including technology, design, implementation, testing etc. Table 9 below show the project risk identified for this project

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Table 9: Technical Risk

Risk ID	Risk	Impact = Possibility x Loss	Key event	Mitigation	Person In Charge
RT01	Security Vulnerabilities/Breach	25	Report of security assessment, security incident	Regular Software Updates, Incident Response Plan, Backup and Recovery, Secure Communication Channels	Software Work Package Manager
RT02	Requirement or Design Change/Flaws	25	During review	Thorough Review, Clear Communication, Prototyping and Iteration, Contingency Planning	Software Work Package Manager
RT03	Approval Workflow Failure	20	Delay or breakdown during approval or validation stages	Clearly defined approval workflow, role-based access control, escalation procedures, workflow automation, and periodic review of approval processes	Software Work Package Manager
RT04	Historical Data Retrieval Failure	15	Failure or delay when accessing archived or legacy data	Data validation checks, standardized data formats, backup and archiving strategy, periodic data integrity testing, and fallback data sources	Software Work Package Manager
RT05	Incorrect Fairness Classification	15	Detection of inconsistent or inaccurate fairness assessment results	Rule validation, test data diversification, periodic model/rule recalibration, audit logging, and peer review of classification logic	Software Work Package Manager
RT06	Host Resources (CPU, Memory) does not Meet Performance Requirement	15	During performance testing	conservative capacity planning assumptions, monitoring system resource utilisation during pilot runs, and committing to execute performance testing prior to full production rollout	Configuration Manager
RT07	Hardware Malfunctions, Device	9	During testing and validation	Quality Equipment, Automated	Configuration Manager

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Risk ID	Risk	Impact = Possibility x Loss	Key event	Mitigation	Person In Charge
	Compatibility Issues			Monitoring, Vendor Support, Thorough Testing and Validation	

3.4 Security

Implementing appropriate security measures is critical to protecting sensitive information, ensuring data integrity, and maintaining the overall reliability of the Task Force Management System. The security measures for this project comprise the following aspects:

1. Human Resource Security

- Ensure that all project team members are authorized and legitimate throughout the project duration.
- Apply role-based access control to ensure that team members can only access information and system functions necessary for their assigned responsibilities.

2. Physical and Environmental Security

- Restrict physical access to project facilities, servers, and equipment to authorized personnel throughout the software development lifecycle.
- Implement physical security controls such as locked cabinets, access cards, and surveillance systems where applicable.

3. Data Security

- Encrypt sensitive data both in transit and at rest to prevent unauthorized access or data leakage.
- Use secure communication channels for data exchange, including encrypted email services and secure file transfer protocols.
- Enforce authentication and authorization mechanisms to protect system access.

4. Documentation and Communication Security

- Project documentation shall be properly maintained and protected by restricting access to both digital and physical copies.
- Secure storage shall be used for sensitive documents, with controlled access to project premises.
- Ensure clear communication within the project team regarding security policies, procedures, and any identified security incidents.

5. Data Backup and Recovery

- Perform regular data backups and securely store backup copies.
- Periodically test data recovery procedures to ensure backups can be restored accurately and within acceptable timeframes.

6. ICT Security Compliance

- All project activities shall comply with applicable ICT security policies, standards, and regulatory requirements.
- Ensure systems and software components are regularly updated with the latest security patches to mitigate known vulnerabilities.
- Encourage prompt reporting of suspicious activities or potential security issues to the project management team for timely action.

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3.5 Interface with associate contractors

Not applicable

3.6 Interface with software IV & V agent(s)

Not applicable

3.7 Subcontractor management

Not applicable

3.8 Formal reviews

Formal reviews will be conducted during software development project as follows:

- Software Specification Review (SSR)
- Preliminary Design Review (PDR)
- Test Readiness Review (TRR)

3.8.1 Review Schedule and Details

Table 10 below shows the scheduled date and objective for Software Specification Review (SSR), Preliminary Design Review (PDR) and Test Readiness Review (TRR):

Table 10: Review Session

Scheduled Date	Review	Objective
16 December 2025	Software Specification Review (SSR)	Establish a clear understanding of the software's features and requirements, setting a baseline for future development.
23 December 2025	Preliminary Design Review (PDR)	Check if the initial design meets the specified requirements and is on the right track.
14 January 2026	Test Readiness Review (TRR)	Ensure all test plans are prepared and that the software is ready for User Acceptance Testing (UAT).

3.8.2 Attendees

Table 11 below shows the participant or attendees that will be gather in the formal review meeting:

Table 11: Reviews Session's Participants

Attendee	SSR	PDR	TRR
Software Package Manager	Required	Required	Required

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Quality Manager	Required	Required	Required
Configuration Manager	Optional	Required	Required
Developer	Required	Required	Required
Software Tester	Optional	Optional	Required
Representative of Client	Required	Required	Required
Cross-Team Manager	Optional	Optional	Optional
Members of other groups	Invited	Invited	Invited

3.8.3 Review Process

1. Preparation:

- Meeting Invitations: Participants will receive meeting invitations **at least one week** prior to the scheduled meeting date. Invitation templates can be found in **Appendix A, B, and C**.
- Document Distribution: The documents to be examined during the reviews are to be submitted **at least 24 hours** before the start of the review.
- Agenda Setting: An agenda outlining the review objectives, discussion topics, and time allocations will be provided.

2. Meeting Planning:

Meetings will be scheduled considering participant availability and project timelines.

3. Conducting the Review:

- The meeting will be chaired by the Software Work Package Manager. One member of the group will take down the minutes.
- Proceedings:
 - Presentation of materials by relevant team members.
 - Open discussions on findings, issues, and concerns.
 - Identification of action items and responsibilities.

4. Report Editing and Issuing:

A comprehensive review report will be prepared, including:

- Summary of discussions.
- List of attendees.
- Decisions made.
- Action items and timelines.

5. Review Pronouncement:

Based on the outcomes, the review will be classified as:

- Adjourned, to be reconducted: If critical issues prevent progression.

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- Accepted with Reservations: If minor issues are present but manageable.
- Accepted: If all criteria are satisfactorily met.

3.8.4 Meeting Location

All formal reviews will be conducted at Multipurpose Room, Level 6, Menara Razak, UTM. For participants who cannot attend in person, secure virtual meeting options will be provided (if necessary).

3.9 Software development library

3.9.1 Software Development Library Structure Overview

The directory structure will be stored in a cloud-based repository, such as OneDrive or Google Drive. In the table 12 below, directory will be the main folder and sub-directory will be the files inside the main folder and "TFMS" refers to the Task Force Management Systems.

Table 12: Library Structure Overview

Directory	Sub-directory	Description
TFMS/SourceCode/		Contains all source code files.
	TaskForceModule	Source code for task force management functionalities.
	SystemIntegrationModule	Source code for other system integration.
	CommonUtilities	Shared libraries and utility functions.
	ThirdPartyLibraries	Externally sourced libraries and their licenses.
TFMS/Documentation/		All project-related documentation.
	Requirements	Requirements documentation.
		Software Requirements Specifications documents.
	Design	Design-related documents.
		Software Design Descriptions and architectural diagrams.
	Testing	Testing documentation
		Comprehensive test planning documents.
		Detailed test case specifications.
		Results from executed test cases.
		Guides and manuals for end-users.

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Directory	Sub-directory	Description
TFMS/Management/		Project management documents.
	ProjectPlans	Project schedules, milestones, and Gantt charts.
	MeetingMinutes	Records of team meetings and action items.
	ProgressReports	Weekly and monthly status reports.
	Review	Documents review for SSR, PDR, TRR.
	TeamContacts	Contact information and roles of team members.
	TeamAttendance	Records of team member attendance, availability, and timesheets.
	KnowledgeSharing	Compile documentation for support group.
TFMS/ConfigurationManagement/		Configuration and change management.
	Baselines	Snapshots of project baselines at key milestones.
	ChangeRequests	Documentation of requested changes and their statuses.
	VersionControl	Logs and tags for version tracking.
TFMS/ToolsAndEnvironments/		Tools and environment configurations.
	DevelopmentTools	Versions/configurations of IDEs, compilers, and debuggers.
	OperatingSystems	Specifications of OS environments used.
	DeploymentScripts	Scripts for building, testing, and deploying the application.
TFMS/IntegrationWithOtherSystems/		Specifics for other system integration.
	Security	Encryption keys and certificates.
	Compliance	Compliance and regulatory documents.
	APIs	API documentation from other system

3.9.2 Access and Control Procedures

Table 13 below shows how the folders and files will be access, secured and backup:

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Table 13: Access and Control Procedure

Aspect	Description
Authentication and Authorization	Secure login credentials with Role-Based Access Control (RBAC).
Audit Trails	Logging all access and modifications; version control histories for changes.
Backup and Recovery	Daily backups at 12pm midnight with monthly integrity tests for data recovery assurance.

3.9.3 Documentation Management

Table 14 below shows how all the documents for the formal reviews will be managed:

Table 14: Documentation Management

Aspect	Description
Collaboration Tools	Use of One Drive or Google Docs for collaborative editing and discussions.
Templates and Standards	Standardized templates ensure consistency across all documents.
Review and Approval Workflows	Documents undergo peer reviews and managerial approvals before baselining.

3.10 Corrective action process

The flowchart below shows a structured approach to dealing with and resolving issues that arise in a project or system, ensuring that issues are properly addressed and resolved. The flowchart depicted in a figure 5 below:

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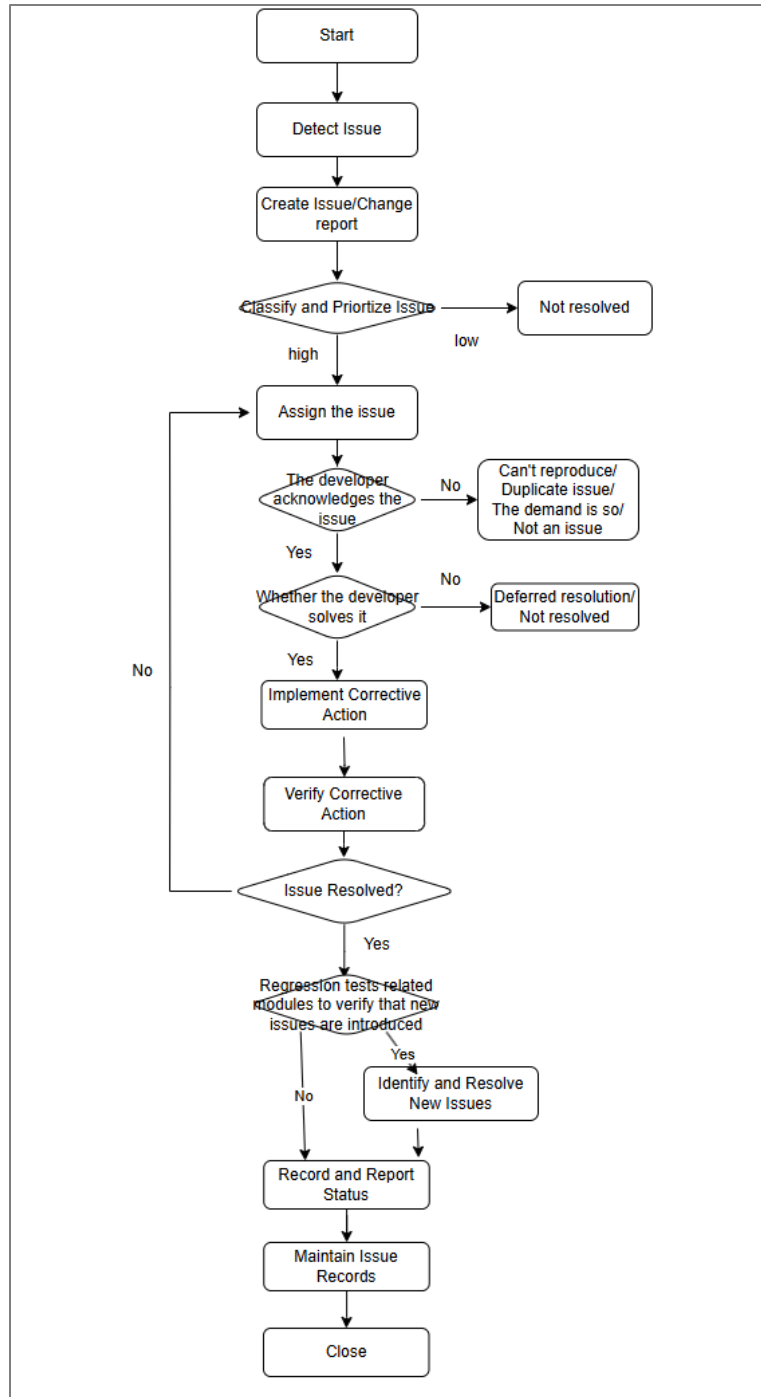


Figure 5: Corrective Flowchart

3.11 Problem/change report

Form is used to record and track problems or changes that occur in a software project. Issue/Change Report form can be found in **Appendix D**.

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4 - SOFTWARE ENGINEERING

This section shall be divided into the following paragraphs and subparagraphs to describe the planning associated with software engineering activities.

The contractor shall perform the software development in compliance with the requirements expressed in this section. Software engineering includes the development team, the tools, documents, methods and procedures working towards software development.

4.1 Organization and resources - Software engineering

This paragraph shall be divided into the following subparagraphs to describe the organization(s) responsible and the resources necessary for software engineering activities.

4.1.1 Organizational structure - Software engineering

The software engineering activities will be performed by six (6) individuals that serve five (5) roles:

Role	Number	Reporting To
Software Work Package Manager	1	Project Manager
Quality Manager	1	Software Work Package Manager
Configuration Manager	1	Software Work Package Manager
Developer	2	Software Work Package Manager
Tester	1	Quality Manager

Table 15: Team Member Roles and Responsibilities

No.	Position	Number	Roles and Responsibilities
1.	Software Work Package Manager	1	- Overall technical leadership - Task allocation and progress tracking - Coordination with Project Manager - Risk management and mitigation
2.	Quality Manager	1	- Define QA processes and standards - Review test plans and results - Ensure compliance with quality requirements - Manage tester and validation activities
3.	Configuration Manager	1	- Version control and repository management - Build and deployment automation - Baseline management and change control

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			Environment configuration
4.	Developer	2	- System design and coding - Unit testing and code reviews - Integration and bug fixing - Documentation of technical components
5.	Tester	1	- Test case design and execution - Defect logging and tracking - Performance and security testing - UAT coordination

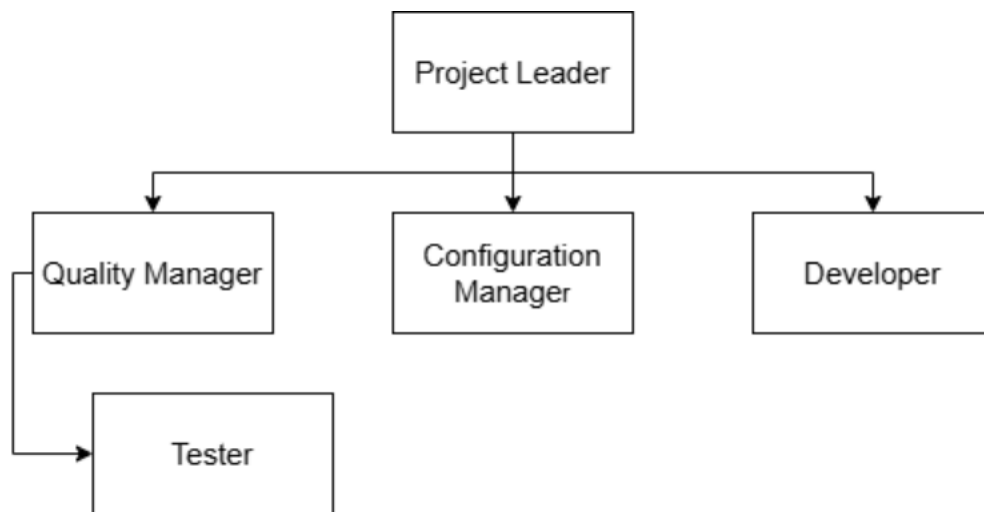


Figure 6: Organizational Structure of the Project

Based on the figure 6 above, a Software Work Package Manager shall manage one (1) Quality Manager, one (1) Configuration Manager, and one (1) Developer during the development process. The Quality Manager shall manage one (1) tester during the testing process.

4.1.2 - Personnel - Software engineering

There are five (5) individuals that are involved in the software development. One (1) person will play the lead role as the software work package manager who manages one (1) quality manager, one (1) configuration manager and one (1) developer. The person who plays the role of the quality manager will manage one (1) tester.

The minimum skill level for each type of position is described in the table below:

Table 16: Skill Level of Personnel

No.	Position	Skills	Minimum Level
-----	----------	--------	---------------

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1.	Software Work Package Manager	Project management	Advanced
		System design	Advanced
		Java	Advanced
		Agile	Advanced
2.	Quality Manager	Quality Assurance	Intermediate
		Testing tools	Intermediate
		Process improvement	Intermediate
3.	Configuration Manager	Version Control Proficiency	Intermediate
		Detail-oriented	Intermediate
		System Management	Intermediate
		Auditing Skills	Intermediate
4.	Developer	Python, Django	Intermediate
		Django REST Framework	Intermediate
		REST APIs	Intermediate
		PostgreSQL	Intermediate
		React, HTML, CSS, JavaScript	Intermediate
5.	Tester	Manual/Automated testing	Intermediate
		Selenium	Intermediate
		Postman	Intermediate
		Test planning	Intermediate

4.1.3 - Software engineering environment

This section describes the plans for establishing and maintaining the software engineering environment, including software, hardware, firmware, and related resources necessary to perform software engineering activities. The environment complies with security-oriented contract requirements, as described in paragraph 3.4 of the Software Development Plan (SDP).

4.1.3.1 - Software items

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The following tables list the software items identified for the software engineering environment:

Table 17: Operating System

Environment	Operating System	Version
Development Environment	Windows 11 Professional	22H2
Integration Environment	Ubuntu Linux	22.04 LTS
Configuration Management Environment	CentOS Stream	8
Staging Environment	Windows 11 Professional	22H2
Production Environment	Windows 11 Professional	22H2

Table 18: Compilers

Compiler	Purpose	Version
Python	Python development	3.11 (Latest stable)

Table 19: Software Tools

Category	Tools	Purpose	Version
Code Auditors and Analyzers	SonarQube	Static code analysis	9.9.0
	Coverity Scan	Dynamic path analysis	Latest stable
Test Drivers and Tools	Selenium WebDriver	Automated testing	Latest stable (4.x)
	Postman	API testing	Latest stable
	JMeter	Performance testing	5.5
Development Tools	Chrome Developer Tools	Troubleshooting	Integrated with Chrome
Documentation Tools	Google Drive	N/A	Automated documentation generation from source code
Database Management System	PostgreSQL	Database development and testing	15.2
Integrated Development Environment (IDE)	Visual Studio Code (VS Code)	Code editing and debugging	1.95

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Security Considerations: The software items listed above are installed on secured systems compliant with ISO 27001 standards. Sensitive data processed in these tools adheres to the data protection policy defined in the SDP.

4.1.3.2 - Hardware and firmware items

The following table 20 list the hardware and firmware items identified for the software engineering environment:

Table 20: Hardware Resources

Category	Device	Specifications
Developer Workstations	Windows 11, 16GB RAM, SSD	Coding and local testing
Test Servers	Linux/Windows, 8GB RAM, 4-core CPU	Integration and system testing
Client-provided Servers	As per Table 3	Staging and production environments

4.1.3.3 - Proprietary nature and Government rights

The Software Licensed identified as below:

- Windows 11 Professional
- PostgreSQL
- SonarQube Enterprise Edition

The Government is granted full rights to use, modify, and distribute any proprietary tools or software specifically developed under this contract, as outlined in Section 3.1 of the SDP.

4.1.3.4 - Installation, control and maintenance

Not applicable

4.2 Software standards and procedures

This paragraph shall be divided into the following subparagraphs to describe the software standards and procedures the contractor plans to use.

4.2.1 - Software development techniques and methodologies

A hybrid methodology combining Sequential and Agile approaches will be used in this system development. Table 21 below is depicted the phases related the methodology:

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Table 21: Development Methodology by Phase

Phase	Methodology	Key Activities
Requirements & Design	Sequential	URS analysis, SRS drafting, system design, SSR/PDR reviews
Development & Testing	Agile (2-week sprints)	Coding, unit testing, integration testing, continuous feedback
Deployment & Handover	Sequential	UAT, deployment, documentation, training

Table 22 below depicts the techniques the contractor plans to use

Table 22: Phases and Techniques

No.	Phases	Techniques
1.	Software Requirements Analysis	- Document Analysis - Interview - Prototyping - Use Cases and Scenarios - Brainstorming
2.	Preliminary Design	- Sketching and Diagramming ✓ System Architecture Diagrams - Modeling ✓ Unified Modeling Language (UML) diagrams ✓ Entity-Relationship (ER) diagrams ✓ Data Flow Diagrams (DFD) - Scenario Analysis
3.	Detailed Design	- Modeling ✓ Unified Modeling Language (UML) diagrams - Entity-Relationship (ER) diagrams ✓ Data Flow Diagrams (DFD) - Scenario Analysis - Design Review Sessions

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No.	Phases	Techniques
4.	Coding and CSU testing	- Coding ✓ System Architecture Diagrams ✓ Modular Programming ✓ Code Reviews ✓ Version Control ✓ Pair Programming ✓ Refactoring ✓ Test-Driven Development (TDD) ✓ Coding Standards ✓ Debugging ✓ Code Optimization - CSU Testing ✓ White Box Testing: Statement Coverage, Branch Coverage, Path Coverage, Condition Coverage ✓ Black Box Testing: Equivalence Partitioning, Boundary Value Analysis (BVA), Decision Table Testing, State Transition Testing, Use Case Testing
5.	CSC integration and testing	- Black Box Testing: Decision Table Testing, State Transition Testing, Use Case Testing - Experience Based Testing: Error Guessing, Exploratory Testing, Checklist - Non-Functional Testing: Security, Performance, Compatibility
6.	CSCI testing	- Black Box Testing: Decision Table Testing, State Transition Testing, Use Case Testing - Experience Based Testing: Error Guessing, Exploratory Testing, Checklist - Non-Functional Testing: Security, Performance, Compatibility, Usability

Table 23 below are the exit criteria for passing condition on to the next phase.

Table 23: Exit Criteria

No.	Phases and Activities	Passing Condition
1.	Requirement Analysis	All functional and non-functional requirements are documented, reviewed, and approved by stakeholders.

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No.	Phases and Activities	Passing Condition
2.	Preliminary Design	Preliminary design is reviewed and approved by stakeholders.
3.	Detailed Design	Detailed system architecture and design documents includes database designs, user interfaces, and system interfaces are documented and approved by stakeholders.
4.	Coding and CSU testing	The system is coded and reviewed according to the design specifications. System passes unit/component testing.
5.	CSC integration and testing	The system passes various levels of testing, including integration testing and system testing. All identified defects must be resolved or accepted as manageable risks. The Test readiness is reviewed and approved by stakeholders.
6.	CSCI testing	The system passes various levels of testing, including user acceptance testing (UAT) and non-functional testing. All identified defects must be resolved or accepted as manageable risks.

The corresponding reviews are described in paragraph 3.8 of the SDP as follow:

- a. Software Specification Review (SSR)
- b. Preliminary Design Review (PDR)
- c. Test Readiness Review (TRR)

4.2.2 - Software development files

Table 24 below are the description of Software Development Files (SDFs) of each Computer Software Unit (CSU), Computer Software Component (CSC) and CSCI.

Table 24: Software Development Files

No.	Software Development Files	Description	File Format
1.	Source Code Files	Contain the actual code written by developers in programming languages such as Python, Java	.py, .java. js
2.	Configuration Files	Store configuration settings and parameters for the application or development environment	.json, .xml, .ini
3.	Build Files	Used to automate the build process, compiling source code into executable applications.	Makefile

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No.	Software Development Files	Description	File Format
4.	Documentation Files	Provide information about the codebase, how to use the software, and instructions for developers	.pdf, .docx, .html
5.	Test Files	Contain test cases and scripts for testing the application to ensure it functions correctly	.test.js (JavaScript test files), .test.py (Python test files), .test.java (Java test files). .docx
6.	Resource Files	Include assets like images, icons, and other multimedia used in the application.	.png, .jpg, .svg, .mp3.
7.	Scripts	Used for automating tasks such as deployment, database migrations, and other repetitive tasks	.sh (Shell scripts), .bat (Batch files),
8.	Data Files	Store data used by the application, such as databases, datasets, or static data.	.csv, .db, .json
9.	Logs Files	Record events and errors that occur during the execution of the application.	.log, .txt
10.	Dependency Files	Specify the external libraries and dependencies required by the application.	package.json, requirements.txt

The location of each SDFs will be referring to paragraph 3.9.1

Maintaining software development files is crucial for ensuring the efficiency, reliability, and consistency of software development process. Table 25 below outlines the structured procedures to effectively maintain SDF files.

Table 25: Maintenance Procedure

No.	Procedure	Processes
1.	Version Control	Use a Version Control Commit Regularly Branching and Merging
2.	Documentation	Code Documentation Technical Documentation
3.	Configuration Management	Update Configuration Files Environment Management

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4.	Build and Deployment Automation	Automated Build Processes Continuous Integration/Continuous Deployment (CI/CD)
5.	Testing	Automated Testing Regular Testing
6.	Backup and Recovery	Schedule Regular Backups Disaster Recovery Plan
7.	Access Control and Security	Implement RBAC Secure Storage
8.	Review and Audits	Conduct Regular Code Reviews and Audits
9.	Housekeeping and Clean Up	Remove Obsolete Files Refactor Code
10.	Collaboration and Communication	Team Collaboration Tools Regular Meeting

4.2.3 Design Standards

Architecture: Layered architecture (Presentation, Business, Data layers)

Design Patterns: MVC, Repository, Service Layer, Dependency Injection

Modeling Tools: Draw.io / Lucidchart for UML diagrams

UML Diagrams Required: Use Case, Class, Sequence, Activity, Deployment

4.2.4 Coding standards

Table 27 shows the coding standards with description that will be implemented

Table 26: Coding Standards

Aspect	Standard
Naming Conventions	• CamelCase for variables/methods • PascalCase for classes • UPPER_SNAKE for constants
Comments	• Javadoc/Pydoc for public APIs

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	• Inline comments for complex logic
Indentation	• 4 spaces (no tabs)
Error Handling	• Structured exception handling with logging
Security	• Input validation • Prepared statements • Encryption for sensitive data
Testing	• Unit test coverage $\geq 80\%$ • Integration tests for key workflows
Code Reviews	• Mandatory peer review before merge

4.3 Non-developmental software

Not Applicable

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5 - FORMAL QUALIFICATION TESTING

This section shall be divided into the following paragraphs and subparagraphs to describe the planning associated with formal qualification testing activities.

5.1 Organization and resources - Formal qualification testing

This paragraph shall be divided into the following subparagraphs to describe the organization(s) responsible and the resources necessary for formal qualification testing.

5.1.1 Organizational structure - Formal qualification testing

Not applicable

5.1.2 Personnel - Formal qualification testing

Not applicable

5.2 Test approach/philosophy

Not applicable

5.3 Test planning assumptions and constraints

Not applicable

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6 - SOFTWARE PRODUCT EVALUATIONS

This section shall be divided into the following paragraphs and subparagraphs to describe the planning associated with software product evaluation activities.

6.1 Organization and resources - Software product evaluations

This paragraph shall be divided into the following subparagraphs to describe the organization(s) responsible and the resources necessary for software product evaluations.

6.1.1 - Organizational structure - Software product evaluations

Not applicable

6.1.2 - Personnel - Software product evaluations

Not applicable

6.2 Software product evaluations procedures and tools

This paragraph shall be divided into the following subparagraphs.

6.2.1 - Procedures

Not applicable

6.2.2 - Tools

Not applicable

6.3 Subcontractor products

Not applicable

6.4 Software product evaluations records

Not applicable

6.5 Activity-dependent product evaluations

Not applicable

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7 - SOFTWARE CONFIGURATION MANAGEMENT

Configuration management consists in applying technical and administrative rules and a supervision in order to: Identify and draw up the functional and physical characteristics of the configuration items.

The developmental configurations of a CSCI are associated with the CSCI design. They are established from the allocated baseline and exist until the product baseline is established. They contain the various states of the SDD and source code. The associated changes do not have to be approved by the Government during the design phase.

This section shall be divided into the following paragraphs and subparagraphs to describe the planning associated with software configuration management (CM) activities.

7.1 Organization and resources - Configuration management

This paragraph describes the organization(s) responsible and the resources necessary for configuration management.

7.1.1 - Organizational structure - Configuration management

Figure 7 shows the overview of the Organizational structure for configuration management.

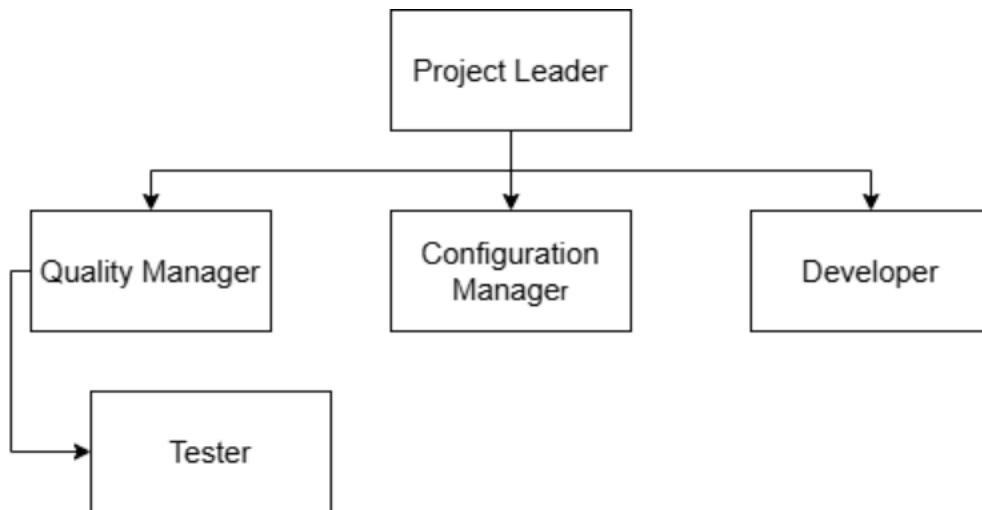


Figure 7: Organizational Structure for Configuration Management

Due to limited team members in this project, the Configuration Manager will work closely with Quality Manager in the Configuration Management

7.1.2 - Personnel - Configuration management

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Position	Responsibilities	Eligibility Requirements
Quality Manager	- Ensures software meets required standards and specifications - Define testing strategies and establish quality assurance protocols - Manages tester for system function verification - Collaborates with configuration manager for software configuration management	- Familiarity with quality management standards (ISO 9001, CMMI). - Familiarity with the software development life cycle - Mastery of quality management tools (e.g. JIRA). - Good leadership and communication skills. - Strong analytical and problem-solving skills.
Configuration Manager	- Responsible for version control, ensuring code, documents, and configurations are accurately tracked and managed - Establish and maintain development environment, ensuring all changes are documented and traceable - Maintain consistency and prevent conflicts during software development and deployment	- Familiarity with version control tools (Git, SVN). - Master automated build tools (Jenkins). - Familiarity with CI/CD processes. - Good documentation and communication skills.

Table 27: Configuration Management Personnel and Skills

7.2 Configuration identification

7.2.1 - Developmental configuration identification

A development configuration is a configuration used to manage and track the evolution of the CSCI (Computer Software Configuration Item) development process. It includes:

- **Software Design Document (SDD):** Describes the detailed design and development process of CSCI.
- **Source code:** Includes the source code after unit testing, and the versions are stored in a version control system to ensure traceability and consistency.
- **Test artifacts:** include test plans, test cases, and test results.

How to establish a configuration in the development phase:

- **The functional baseline:** corresponds to the reviewed system requirements.
 - Testing

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- Library
- Code
- Bugs
- **The allocated baseline:** corresponds to the reviewed software requirements specification and software interface requirements specification.
 - Specification
 - Design
 - Management Plan
- **The product baseline:** corresponds to the completed software product delivered for system integration.
 - User Manual
 - Supporting Documents:
 - Meeting Minutes
 - Problem Change Request (PCR) Form

7.2.2 - Identification methods

Identification Method:

- **Naming and tagging:** All configuration items, including documentation and code, will follow uniform naming and tagging rules to ensure easy identification and reference throughout the project.
eg:

System identification: SYSNAME-Company-YYYY-SS
CSCI identification: CSCINAME-Company-YYYY-SS
- **Version control:** Use a version control system, such as Git, to manage the versions of all configuration items, ensuring that all changes are recorded and approved.
- **Document control:** All project documents will be tagged and controlled to ensure that the version of the document is consistent with the version of the software.

In this project, configuration items (CIs) are identified and controlled at multiple levels to ensure effective traceability, change management, and impact analysis. The CI hierarchy follows a structured decomposition approach as described below:

Computer Software Configuration Item (CSCI):

- A CSCI represents a major deliverable software element that is independently managed and baselined.
- Each CSCI typically corresponds to a subsystem or major functional module defined in the Software Design Description (SDD).
- Examples include authentication module, reporting module, or data management module.

Computer Software Component (CSC)

- A CSC is a logical subdivision of a CSCI, representing a cohesive set of related software functions.
- CSCs are used to structure the design and facilitate modular development and testing.
- Each CSC is uniquely numbered within its parent CSCI.

Computer Software Unit (CSU)

- A CSU is the lowest-level software element, typically corresponding to a source code file, class, or executable unit.

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- CSUs are directly implemented in source code and are the primary units of change and version control.

The hierarchical relationship is: **System → CSCI → CSC → CSU**

In order to track revisions to documentation and CSU, CSC, CSCI, the following methods are used in this project:

- **Change logging/documentation:**
Each revision needs to be recorded in the changelog and documentation, including the revision date/version number/revision reason/revision initiator and approver.
- **Versioning:**
Use a version control system (e.g. Git) to manage all revisions, ensuring that every commit is documented and can be traced back to the specific committer and reason for the change.

Below is a configuration identification format applying this project:

- **System Identification:**

TFMS-CODEForce

Where:

TFMS = System Name
CODEForce = Company Short Name

- **Document Naming Convention (DOCNAME):**

XXX-TFMS-DDMMYY-VX

Where:

XXX = Document type (e.g., SDD, SDP, SRS, STD)
TFMS = System Name
DDMMYY = Revision index (Date Format: DDMMYY)
VX = Version index (e.g., V1, V2)

Example:

SDD : SDD-TFMS-171225-V1
SDP : SDP-TFMS-171225-V1
SRS : SRS-TFMS-171225-V1
STD : STD-TFMS-171215-V1

- **The numbering for the Source Data File (SDF):**

TFMS-01-100-FFF-X.Y

Where:

TFMS = System name

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01	= Functional module / subsystem number
100	= Source Code
FFF	= File type (e.g., CPP, BAS, DAT, TXT, HTM)
X.Y	= CSC number (X) and CSU number (Y)
X	= Refer to CSC
Y	= Refer to CSU

7.3 Configuration control

This paragraph shall be divided into the following subparagraphs to provide a detailed description of the procedures to be used in controlling changes to and maintaining the Developmental Configuration(s) and internally controlled documentation.

7.3.1 - Flow of configuration control

The figure 7 below shows the configuration control flowchart.

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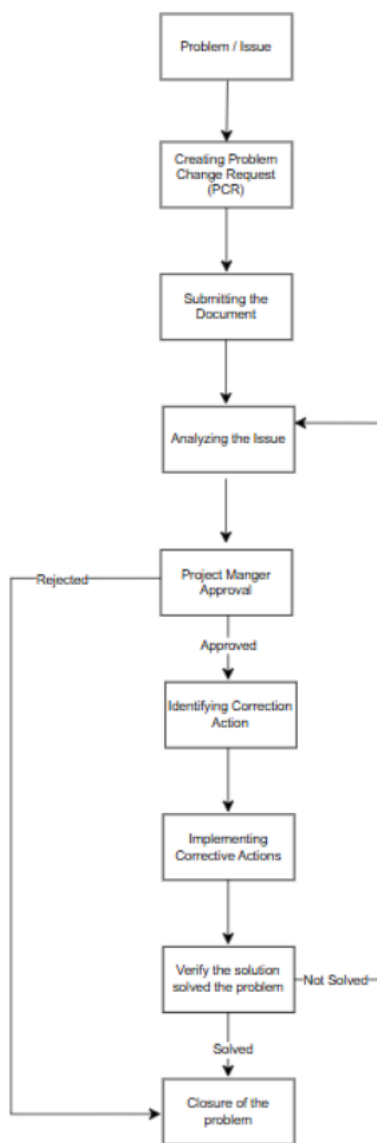


Figure 8: Configuration Control Flowchart

7.3.2 - Reporting documentation

This subparagraph shall be divided into the following subparagraphs to describe or reference the description of the reporting documentation to be used in controlling software problems and changes.

7.3.2.1 - Problem Change Report (PCR)

The Problem Change Report (PCR) form is used to document any configuration changes that are made during the development process of the software which will be approved or rejected by the project manager.

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Project Name					
Date of request submission					
Title of change Request					
Submitted by					
Change No.					
Change Category	Scope	Schedule	Cost	Technology	Other
Priority	High		Medium		Low
Change Description					
Trigger event that made the change necessary					
Change reason					
Assigned by					

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Duration				
Impact of the change on:				
Scope				
Cost				
Labor				
Schedule				
Risk				
Other				
Problem status				
Solution				
Approval Status	Approved	Rejected	Waiting	

7.3.3 - Review procedures

This section outlines the purpose and procedures for review boards responsible for configuration control.

7.3.3.1 - Configuration Control Board (CCB) Procedures

The figure 8 below shows the review procedure.

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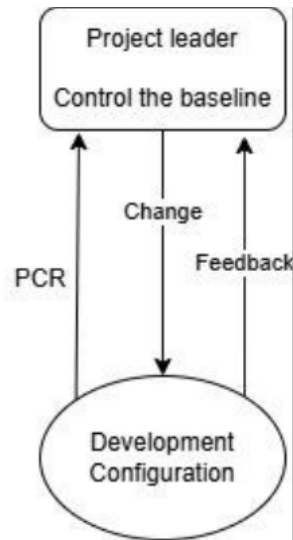


Figure 9: Review Procedures

Purpose:

The CCB evaluates and approves or rejects all Problem Change Reports (PCR) to ensure alignment with baselines and project goals.

Procedures:

1. Submission: PCR are submitted and logged into the Configuration Control Board (CCB) for tracking and review.

2. Review Meetings:

Evaluate technical, cost, and schedule impacts.

Decisions: Approved, Rejected, or Deferred.

3. Post-Review: Approved changes are implemented, recorded in the CCB, and baselines are updated.

Traceability:

All decisions are documented to maintain historical records and alignment with the configuration identification scheme.

7.3.4 - Storage, handling, and delivery of project media

This section outlines the procedures for the secure storage, handling, and delivery of project media, including software, documentation, and other configuration items.

1. Storage:

- All project media (software, documentation, and configurations) are stored in a secure version-controlled repository (e.g., GitLab, Subversion).
- Backups:
 - Weekly backups are created and stored at an offsite location.
 - Redundant copies are maintained to ensure recovery in case of data loss.
- Access to the storage system is restricted to authorized personnel and is monitored through audit logs.

2. Handling:

- Access Control:
 - Only personnel with appropriate permissions may retrieve or modify stored media.
 - Role-based access ensures that changes are limited to approved individuals.

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- Logging and Tracking:
 - All accesses and modifications are logged, including timestamps, user IDs, and actions taken.
 -

3. Delivery:

Deliverable software and documentation are encrypted before transfer to ensure confidentiality.

- Delivery methods include:
 - Digital Delivery: Secure channels such as SFTP or secure cloud platforms.
 - Physical Media: Labeled and tracked USB drives, DVDs, or other secure storage devices.

4. Verification:

A delivery checklist is used to confirm that all required items are included. Recipients acknowledge receipt and verify content integrity.

The check-in and check-out process is defined in Figure 9 and 10 below.

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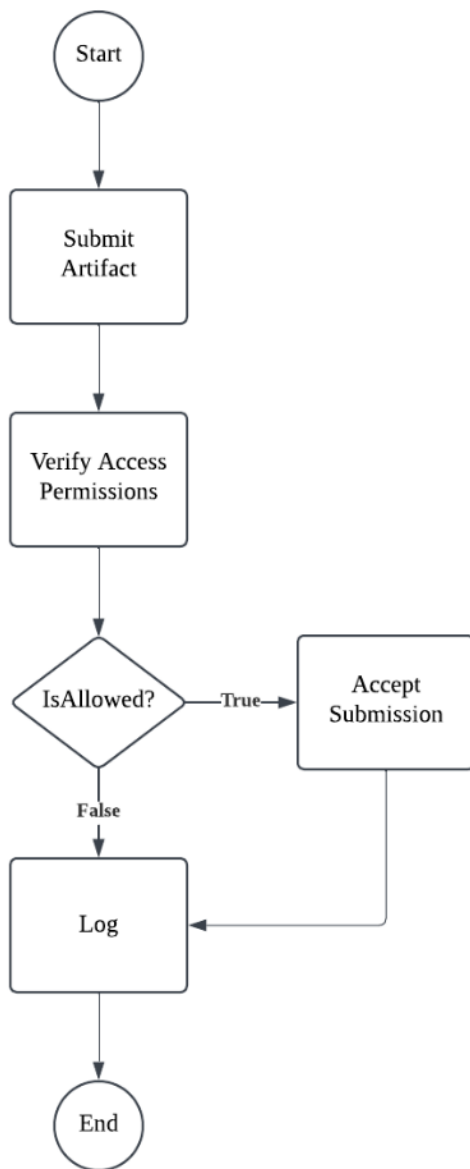


Figure 10: Check-in Artifact Process Flow

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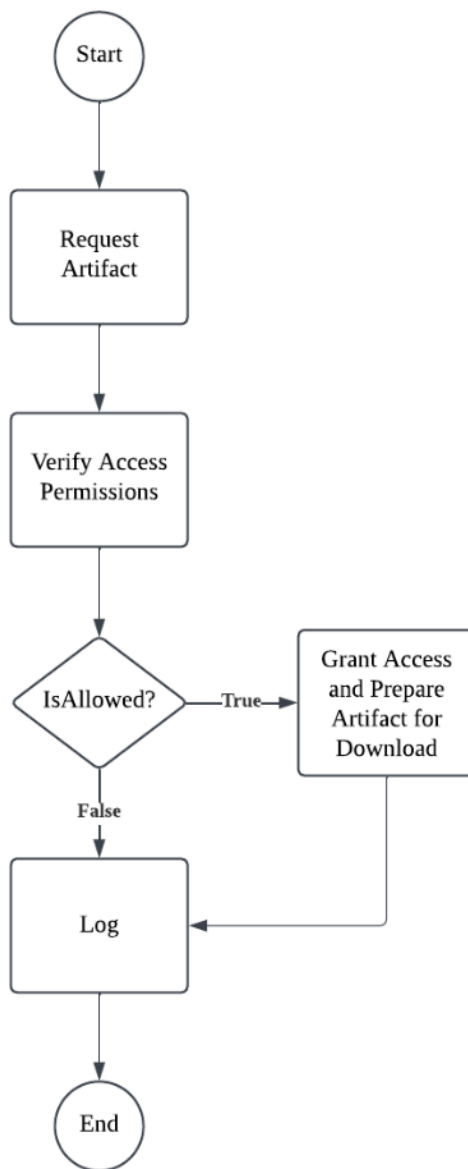


Figure 11: Check-out Artifact Process Flow

7.3.5 - Additional control

Not Applicable

7.4 Configuration status accounting

7.4.1 Overview of Configuration Status Accounting

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The configuration status accounting system provides visibility into the status of all configuration items and changes through:

- Record Configuration Items: Identify all software components, interfaces, and documents with unique IDs and version numbers.
- Track Changes: Monitor change requests, problems, and enhancements.
- Maintain Version Histories: Keep records of software and document versions.
- Provide Reports: Show configuration status and changes during development.

7.4.2 Content of Status Accounting Records

1. List of Configuration Items.

Table 28 shows a preliminary list of software components and documents to be developed with a unique identifier for each. DOC-001 stands for “Document 001” and CI-001 stands for “Configuration Item 001”.

Example format:

Table 28: Configuration Item Register

CI ID	Item Name	Type	Current Version	Status	Owner
CI-001	Backend	Software	1.0.0	Under Development	Lead Developer
CI-002	Frontend	Software	1.0.0	Under Development	Frontend Developer
CI-003	Database Schema	Data	1.0.0	Baseline Established	Database Developer
CI-004	SRS Document	Documentation	1.0	Approved	Quality Manager
CI-005	SDD Document	Documentation	0.5	Draft	Lead Developer

2. Change Request Status

Table 29 shows the process for recording and tracking change requests if there is any. CR-001 stands for “Change Request 001”.

Example format:

Table 29: Change Request Status Process

PCR ID	Description	Status	Submitted	Approved	Implemented
PCR-001	Add export to PDF feature	Approved	2025-12-10	2025-12-12	2025-12-15
PCR-002	Fix login timeout issue	In Progress	2025-12-15	2025-12-16	TBD

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PCR-003	Enhance report filtering	Pending Review	2025-12-18	TBD	TBD
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3. Approved Changes Implementation

Table 30 shows procedures for documenting approved changes and implementation status. ECP-001 stands for "Engineering Change Proposal 001".

Example format:

Table 30: Approved Changes Process

Change ID	Description	Implementation Status	Impacted CIs
ECP-001	Enhance security protocols	Pending	CI-001, CI-003

4. Version and Distribution History

Table 31 shows the records and tracks of version history of the software and associated documentation.

Example format:

Table 31: Version History

Version	Description	Date Released	Distributed To	Remarks
1.0	Initial release	YYYY-MM-DD	Development Team	Baseline version

7.4.3 Configuration Status Accounting Reports

Planned reports include:

1. Configuration Index Report

Table 31 shows a list of configuration items, including their versions and changes, tracked from the planning phase through the development process.

Example format:

Table 31: Configuration Index Report

Configuration Item	Description	Current Version	Included Changes	Change Date	Status	Responsible Team Member
CI-001	Banking API Integration	v1.2	Enhanced security protocols	YYYY-MM-DD	Approved	Amirul

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2. Change Status Report

Table 32 shows the detail of all change request and their statuses. PCR-001 stands for “Project Change Request 001”

Example format:

Table 32: Change Status Report

Change Request ID	Description	Status
PCR-001	Fix transaction timeout issue	Pending

3. Document Status Report

Table 33 shows the current status of all project documents, including revisions and updates.

Example format:

Table 33: Document Status Report

Document	Version	Status	Last Updated
Software Requirements Specification	0.1	Draft	TBD

4. Release Notes

Table 34 shows summarization of the changes and updates included in each software release as they occur.

Example format:

Table 34: Release Notes Report

Release Version	Release Date	Description	Key Updates
TBD	TBD	Initial release of the vending machine system with banking integration	Basic transaction functionality

7.4.4 Reporting Schedule

Table 34 shows reporting schedule.

Table 35: Reporting Schedule

Report Type	Frequency	Purpose	Recipients
Weekly Status Report	Weekly	Project Team, Client	Track progress and issues
Change Status Report	Bi-weekly	CCB Members	Monitor change implementation

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Release Notes	Per Release	All Stakeholders	Document release contents
Audit Report	Monthly	Quality Manager, Client	Compliance verification

7.5 Configuration audits

7.5.1 Functional Configuration Audit (FCA)

Process:

1. Review test documents to check compliance with requirements.
2. Verify that all approved changes are included.
3. Confirm all approved changes are incorporated
4. Validate system functionality against requirements

Documentation Needed: Complete specifications of the software component.

For details of the FCA checklist, refer to **Appendix F**.

7.5.2 Physical Configuration Audit (PCA)

Process:

1. Audit design documents such as the Software Design Description (SDD)
2. Verify that acceptance test requirements are met.
3. Check completeness of deliverables
4. Validate deployment artifacts

Post-Audit Changes: Any changes after the PCA require a formal change proposal.

For details of the PCA checklist, refer to **Appendix G**.

7.5.3 Audit Schedule and Responsibilities

Table 36 shows audit schedule.

Table 36: Audit Schedule

Audit Type	Timing	Objectives
Functional Configuration Audit (FCA)	Before accepting each software component	Verify functional compliance

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Physical Configuration Audit (PCA)	After the FCA and before finalising product	Verify physical configuration
Interim Audits	Monthly or as needed	Ongoing compliance check

Table 37 shows responsibility for each role.

Table 37: Responsibilities

Roles	Responsibilities
Configuration Manager	Prepare audit materials, conduct audit, document findings.
Quality Manager	Review audit process, ensure compliance, track corrective actions.
External Auditor	Independent verification and validation.
Project Manager	Ensure team readiness, oversee corrective actions.

7.5.4 Deliverables

After audits are conducted, the following will be provided:

- Audit Reports: Findings from the audits.
- Updated Configuration Records: Reflecting any changes identified.
- Revised Documentation: Updates based on audit findings.

7.6 Preparation for specification authentication

The preparation for specification authentication in SCM is critical for ensuring the accuracy, consistency, and quality of software products. It facilitates effective change management, collaboration, compliance, and project management, to ensure more efficient and successful software development processes.

According to SWEBOK, the key steps are as follows:

1. Submitting Specifications to the Contracting Agency for Review and Authentication

- **Purpose:** Establish a formal agreement on the accuracy and completeness of the specifications.
- **Process:**
 - **Specification Compilation:** Collect all relevant specifications, ensuring alignment with the agreed functional and non-functional requirements.
 - **Internal Validation:** Perform internal reviews to confirm adherence to established standards and detect potential inconsistencies.
 - **Formal Submission:** Submit the specifications, supported by documentation such as traceability matrices and compliance checklists, to the contracting agency for review.
 - **Feedback Integration:** Address feedback from the contracting agency to ensure the specifications meet their requirements.

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2. Ensuring the Incorporation of Approved Changes

- **Purpose:** Maintain the integrity of the baseline by incorporating only authorized changes.
- **Process:**
 - **Change Approval:** Evaluate change requests through a defined **Change Control Process** to assess their impact on the current baseline.
 - **Controlled Updates:** Apply the approved changes to the specifications while maintaining compliance with the authenticated baseline.
 - **Change Validation:** Conduct rigorous testing and verification to ensure that the changes do not introduce errors or inconsistencies.
 - **Stakeholder Notification:** Communicate changes to relevant stakeholders, including development, testing, and deployment teams.

3. Updating Configuration Status Accounting Reports to Reflect Approved Baselines

- **Purpose:** Provide an accurate and auditable record of the configuration status of specifications.
- **Process:**
 - **Baseline Adjustment:** Update the baseline in the **Configuration Management System (CMS)** to reflect changes approved by the contracting agency.
 - **Status Accounting Reports:** Generate updated **Configuration Status Accounting Reports (CSARs)** to detail the current state of all configuration items (CIs), including specifications.
 - **Stakeholder Dissemination:** Share the updated reports with stakeholders to ensure transparency and alignment across teams.
 - **Audit and Verification:** Perform audits to validate that the reports accurately represent the current configuration and comply with project objectives.

Below are the key roles for this process:

- The authentication is the customer's responsibility and is performed on the requirement specification documents.
- The contractor approves the requirement specifications for execution.

7.7 Configuration management major milestones

7.7.1 Configuration Identification

- **Objective:** Establish the baseline for configuration items (CIs) to enable tracking and control.
- **Activities:**
 - Identifying configuration items (e.g., code, requirements, design documents).
 - Defining naming conventions and unique identifiers for each CI.
 - Establishing relationships between configuration items.
- **Milestone:** Completion of the configuration identification phase, where all necessary items are documented and organized into the system.

7.7.2 Configuration Baseline Establishment

- **Objective:** Define and formalize baselines at critical stages of the development lifecycle to serve as points of reference for change.
- **Types of Baselines:**
 - **Functional Baseline:** Initial baseline that captures system-level requirements.
 - **Allocated Baseline:** Represents the assignment of functionality to subsystems or components.

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- **Product Baseline:** Reflects the completed and verified system design, including implementation details.
- **Milestone:** Baseline approval ensures consistency for subsequent changes.

7.7.3 Change Control Milestone

- **Objective:** Control modifications to the baseline and prevent unauthorized changes.
- **Activities:**
 - Submitting and reviewing change requests.
 - Evaluating the impact of proposed changes on cost, schedule, and quality.
 - Approving or rejecting changes via the Change Control Board (CCB).
- **Milestone:** Approval of change requests and initiation of modifications.

7.7.4 Configuration Status Accounting

- **Objective:** Provide visibility into the current status of configuration items, including changes and updates.
- **Activities:**
 - Maintaining accurate records of configuration items.
 - Documenting changes to baselines and their implementation status.
- **Milestone:** Updated status accounting reports that reflect current configuration and changes.

7.7.5 Configuration Audits

- **Objective:** Verify the integrity of baselines and ensure compliance with requirements.
- **Types of Audits:**
 - **Functional Configuration Audit (FCA):** Confirms that the system meets its functional requirements.
 - **Physical Configuration Audit (PCA):** Verifies that the physical implementation matches the documentation and design.
- **Milestone:** Completion of audits confirms configuration integrity and readiness for release.

7.7.6 Release Management and Delivery

- **Objective:** Ensure that the system is properly packaged, controlled, and delivered to stakeholders.
- **Activities:**
 - Assembling and verifying all components of the release.
 - Creating version-controlled builds.
- **Milestone:** Delivery of a stable, approved product version to the end-user or client.

7.7.7 Archiving and Closure

- **Objective:** Finalize configuration management activities by archiving project documentation and baselines.
- **Activities:**
 - Storing records for future reference.
 - Formal closure of configuration control activities.
- **Milestone:**
 - Baseline establishment.
 - Change control implementation.
 - Audit completion.
 - Final delivery.

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8 - OTHER SOFTWARE DEVELOPMENT FUNCTIONS

This section shall be divided into the following paragraphs and subparagraphs to describe any other contractor functions involved in the software development effort.

These other activities can be, for instance: the software centralized archiving, the development of a simulation environment, the documentation plan.

8.X -Function name

This paragraph (beginning with 8.1) shall be associated with any function.

This paragraph shall describe the function to be performed.

This paragraph shall be divided into the following subparagraphs to describe the organizational structure, resources, and the methods and procedures necessary to perform the function.

8.X.1 -Organizational structure - (function name)

This subparagraph shall describe the organization(s) responsible for performing the function.

This subparagraph shall include the authority and responsibilities of each organization and its relationship to other organizational entities such as the organization(s) responsible for performing configuration management.

If more than one organization is involved, the precise structure, personnel, and resources of each organization and their interrelationship shall be highlighted.

For instance: team flowchart.

8.X.2 -Personnel - (function name)

This subparagraph shall describe the number and skill levels of personnel who will perform the function.

The personnel shall be described by title and minimum qualifications for the position.

In addition, this subparagraph shall specify any requirement unique to particular positions, such as geographic location, security level, extended hours, etc.

The names of individuals shall be given only given if demanded in the contract.

For instance: Estimated and planned personnel number. The requirements for training the team to a language, to a method or to the field of application of the project may be evaluated here.

8.X.3 -Other resources - (function name)

This subparagraph shall identify and describe any other resources necessary for performing the function.

For each resource, this subparagraph shall briefly describe the aspect of the function that requires the resource.

8.X.4 -Methods and procedures - (function name)

This subparagraph shall describe the methods and procedures to be used to perform the function.

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9 - NOTES

This section shall contain any general information that aids in understanding this document.

CSCI – Computer Software Configuration Item
CCB – Configuration Control Board
FCA – Functional Configuration Audit
PCA – Physical Configuration Audit
PCR – Problem Change Report
PDR – Preliminary Design Review
SDD – Software Design Description
SDP – Software Development Plan
SRS – Software Requirements Specification
SSR – Software Specification Review
TFNS – Taskforce Monitoring System
TRR – Test Readiness Review
URS – User Requirements Specification

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